

Generation Interconnection System Impact Study Report

For Queue Project Q21-1910

Straits 138 kV
September 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1910. The Interconnection Customer proposes a 280.6 MW battery energy storage facility interconnecting at the Straits 138 kV substation in the Virginia area. The request was received on October 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 683 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$36,331,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DELTAPORT SUBSTATION / POTOSI 138.0 kV	SWITZER / PHEASANT BRANCH SUBSTATION 138.0 kV	131.8	365	5.6
POTOSI / TYLER 138.0 kV	PHEASANT BRANCH / SWITZER 138.0 kV	129.4	396	5.6
WELLS IC / SANDY VLY 138.0 kV Line	BEAR / MECKLENBURG POWER STA 345.0 kV	125.7	243	7.1
DELTAPORT SUBSTATION / MOONSTONE 138.0 kV	SWITZER SUBSTATION / PHEASANT BRANCH 138.0 kV Line	125.7	352	5.6
CITY W DIESEL PLANT SUBSTATION / ROBERTS SW STA SUBSTATION 345.0 kV line	NAUVOO SUBSTATION / ROBERTS SW STA SUBSTATION 345.0 kV	118.8	1,471	7.1
BAIROA 3008 / BANK OF AMERICA SUBSTATION 138.0 kV	HAGGIN / N TOWER TAP 345.0 kV	118.7	315	6.7
OLD HICKORY SUBSTATION / DEWALT 138.0 kV	NORTH TOWER TAP / DPC ELROY 138.0 kV	117.4	517	22.7
GENEVA BORDER CITY SUBSTATION / KEOKUK 138.0 kV L	DPC ELROY / N TOWER TAP 138.0 kV	117.2	426	44.2
GENEVA BORDER CITY / DEWALT 138.0 kV	DPC ELRY / NORTH TOWER TP 138.0 kV	116.6	535	22.7
GENEVA BORDER CITY / WELLS IC 138.0 kV	BEAR / MECKLENBURG POWER STATION 345.0 kV	115.8	435	13.3
SYL LASKIN / MITCHELL DAM 138.0 kV	HAGGIN / MITCHELL DAM SUBSTATION 345.0 kV	115.7	350	6.5
WING LANE / SECO 138.0 kV Line	MITCHELL DAM / SKAGGS ISLAND JUNCTION 2 345.0 kV	113.3	573	9.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SOLANO IRRIGATION DIST / N FOND DU LAC 138.0 kV	WELLS IC SUBSTATION / SANDY VLY 138.0 kV L	113.0	240	11.3
BEAR ISLAND 345-1 kV Autotransformer	BEAR ILAND / CHANCELLOR ROW 345.0 kV L	112.8	588	7.1
DPC ELROY / NORTH TOWER TAP 138.0 kV	MCCARTNEY / NORTH TOWER TAP 345.0 kV line	112.2	853	48.6
MITCHELL DAM 345-138 kV XFMR	HAGGIN SUBSTATION / MITCHELL DAM 345.0 kV	111.5	363	6.5
BRANDYWINE POWER FACILITY SUBSTATION / ROBERTS SW STA 345.0 kV	GLENANNAN / ROBERTS SW STA 345.0 kV L	111.3	1,300	6.2
WESTCOAST CELLUFIBRE / LEAVITT 138.0 kV #1	SKAGGS ISLAND JCT 2 / MITCHELL DAM 345.0 kV	110.9	561	8.9
WESTCOAST CELLUFIBRE / LEAVITT 138.0 kV Line 2	SKAGGS ISLAND JUNCTION 2 SUBSTATION / MITCHELL DAM 345.0 kV	110.9	561	8.9
REDBED PLAINS WIND FARM / FRUITRIDGE & S. LAND PARK SUBSTATION 138.0 kV	MITCHELL DAM SUBSTATION / SKAGGS ISLAND JUNCTION 2 345.0 kV Line	110.5	444	7.1
SOLANO IRRIGATION DIT / WELLS IC 138.0 kV	SANDY VLY / WELLS IC 138.0 kV	110.4	237	11.3
N TOWER TAP SUBSTATION / HAGGIN 345.0 kV	BEAR / MECKLENBURG POWER STA 345.0 kV line	110.2	1,742	18.4
BEAR / FLUVANNA P.S 345.0 kV	BIRCHWOOD / LEAVITT 345.0 kV L	110.0	2,382	17.6
OYSTER RIV / CEDAR FALLS WI 138.0 kV line	N TOWER TAP / HAGGIN 345.0 kV	109.9	212	7.3
HARVELL / OYSTER RIVER 138.0 kV	N TOWER TAP SUBSTATION / HAGGIN SUBSTATION 345.0 kV	109.9	191	7.3
N RANDOLPH / BAIROA 3008 138.0 kV	HAGGIN / N TOWER TAP 345.0 kV	109.0	323	6.7
HAVARD / SECO 138.0 kV	MITCHELL DAM SUBSTATION / SKAGGS ISLAND JUNCTION 2 345.0 kV L	108.4	705	9.9
CLOVER / BEAR 345.0 kV	MECKLENBURG POWER STA / BEAR 345.0 kV	108.2	4,332	11.0
FRUITRIDGE & S. LAND PARK / CARLIN SUBSTATION 138.0 kV	SKAGGS ISLAND JUNCTION 2 / MITCHELL DAM 345.0 kV	108.0	495	7.1
CAPE HORN / YATES DAM 138.0 kV	HUTCHINS / BAUXITE SUBSTATION 138.0 kV	106.0	169	8.2
RIVERVIEW / PERSON 345.0 kV	LEESVILLE / CLOVER 345.0 kV line	103.5	1,654	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DELTAPORT SUBSTATION / POTOSI 138.0 kV	Rebuild line with higher-capacity conductor	\$6,514,000	4% (\$244,000)
POTOSI / TYLER 138.0 kV	Rebuild line with higher-capacity conductor	\$536,000	4% (\$20,000)
WLLS IC / SANDY VALLEY 138.0 kV	Rebuild line with higher-capacity conductor	\$16,651,000	33% (\$5,558,000)
DELTAPORT / MOONSTONE 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,396,000	4% (\$90,000)
CITY WEST DIESEL PLANT / ROBERTS SW STA 345.0 kV line	Rebuild line with higher-capacity conductor	\$7,281,000	2% (\$124,000)
BAIROA 3008 SUBSTATION / BANK OF AMERICA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,217,000	6% (\$137,000)
DEWALT / OLD HICKORY 138.0 kV	Rebuild line with higher-capacity conductor	\$5,610,000	28% (\$1,572,000)
GENEVA BORDER CITY / KEOKUK 138.0 kV	Rebuild line with higher-capacity conductor	\$7,761,000	35% (\$2,686,000)
GENEVA BORDER CITY / DEWALT 138.0 kV	Rebuild line with higher-capacity conductor	\$20,945,000	28% (\$5,868,000)
GENEVA BORDER CITY / WELLS IC 138.0 kV	Rebuild line with higher-capacity conductor	\$10,125,000	27% (\$2,735,000)
SYL LASKIN / MITCHELL DAM 138.0 kV	Rebuild line with higher-capacity conductor	\$1,770,000	3% (\$50,000)
WING LANE / SECO 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,944,000	3% (\$61,000)
SOLANO IRRIGATION DIST / N FOND DU LAC 138.0 kV	Rebuild line with higher-capacity conductor	\$17,283,000	31% (\$5,411,000)
BEAR SLAND 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$17,676,000	2% (\$420,000)
DPC ELROY / NORTH TOWER TAP 138.0 kV	Rebuild line with higher-capacity conductor	\$6,000,000	24% (\$1,458,000)
MITCHELL DAM 345-138 kV XFMR	Replace transformer with higher-rated unit	\$14,966,000	3% (\$423,000)
ROBERTS SW STA / BRANDYWINE POWER FACILITY 345.0 kV L	Rebuild line with higher-capacity conductor	\$26,096,000	2% (\$524,000)
WESTCOAST CELLUFIBRE / LEAVITT 138.0 kV #1	Rebuild line with higher-capacity conductor	\$1,511,000	3% (\$40,000)
WESTCOAST CELLUFIBRE / LEAVITT 138.0 kV Line 2	Rebuild line with higher-capacity conductor	\$1,453,000	3% (\$38,000)
REDBED PLAINS WIND FARM / FRUITRIDGE & S. LAND PARK SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$8,894,000	3% (\$246,000)
SOLANO IRRIGATION DIT / WELLS IC 138.0 kV	Rebuild line with higher-capacity conductor	\$6,503,000	31% (\$2,036,000)
N TOWER TAP SUBSTATION / HAGGIN 345.0 kV	Rebuild line with higher-capacity conductor	\$53,554,000	3% (\$1,397,000)
FLUVANNA P.S / BEAR 345.0 kV line	Reconductor line	\$40,514,000	2% (\$743,000)
CEDAR FALLS WI / OYSTER RIVER 138.0 kV line	Reconductor line	\$839,000	7% (\$56,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HARVELL / OYSTER RIVER 138.0 kV	Reconductor line	\$931,000	7% (\$62,000)
N RANDOLPH / BAIROA 3008 138.0 kV	Reconductor line	\$2,261,000	6% (\$147,000)
HAVARD / SECO 138.0 kV	Reconductor line	\$1,723,000	3% (\$54,000)
CLOVER / BEAR 345.0 kV	Reconductor line	\$44,933,000	1% (\$577,000)
CARLIN / FRUITRIDGE & S. LAND PARK 138.0 kV	Reconductor line	\$587,000	3% (\$16,000)
CAPE HORN / YATES DAM 138.0 kV	Reconductor line	\$8,638,000	36% (\$3,118,000)
RIVERVIEW / PERSON 345.0 kV	Reconductor line	\$31,106,000	1% (\$420,000)
	Total allocated to project		\$36,331,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.